A Sharp Two-Point Counterexample to a Dirichlet-Series Pick Converse

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Status: candidate counterexample, likely valid; human review needed. Simultaneous positivity of the two necessary Pick matrices in Chavan–Sahu's Theorem 2.1 does not force a contractive Dirichlet-series interpolant. An explicit rational data set gives a full negative answer to the question in their concluding remarks.

1 Source question

Sameer Chavan and Chaman Kumar Sahu study contractive interpolation by functions in $H^\infty(\mathbb{H}_0)\cap\mathcal{D}$, where $\mathcal{D}$ denotes the functions representable by a convergent Dirichlet series in some right half-plane [1]. Their Theorem 2.1 says that interpolation data $\lambda_1,\dots,\lambda_n\in\mathbb{H}_{1/2}$ and $w_1,\dots,w_n\in\mathbb{D}$ must satisfy

$$P_\zeta=\left[(1-\overline{w}_iw_j)\zeta(\overline{\lambda}_i+\lambda_j)\right]_{i,j=1}^n\succeq 0, \tag{1}$$

$$P_{\mathrm{cl}}=\left[\frac{1-\overline{w}_iw_j}{\overline{\lambda}_i+\lambda_j}\right]_{i,j=1}^n\succeq 0,\qquad \operatorname{rank}P_{\mathrm{cl}}=n. \tag{2}$$

Condition (2) makes the classical contractive $H^\infty(\mathbb{H}_0)$ solution family $\mathcal{S}$ non-singleton. On PDF page 8, the authors ask whether, if (1) also holds, there must be a solution in $\mathcal{S}\cap\mathcal{D}$.

Concluding remarks

We conclude this paper with an observation towards a possible converse of Theorem 2.1. Suppose that Theorem 2.1(ii) holds. Then, by Lemma 2.1, the interpolation problem

$$\varphi(\lambda_i)=w_i,\quad i=1,2,\dots,n \text{ with } \varphi\in H_1^\infty(\mathbb{H}_0)$$

admits more than one solutions. Hence, by an application of [**3**, Corollary 8.82], the set of all solutions φ is of the form

$$\mathcal{S}:=\{\tilde{g}_{1,1}+\tilde{g}_{1,2}\tilde{g}_{2,1}(1-\tilde{g}_{2,2}h)^{-1}:h\in H_1^\infty(\mathbb{H}_0)\},$$

where $\tilde{g}_{i,j}=g_{i,j}\circ C$ (see (2.1)), each $g_{i,j}$ is a rational function of degree at most n with poles outside the closed unit disc, $\tilde{g}_{1,2}(\lambda_i)=0$ for all $i=1,2,\dots,n$, and the matrix $[g_{i,j}]_{i,j=1}^2$ is unitary on the unit circle $\mathbb{T}$. However, even if in addition Theorem 2.1(i) holds, we do not know whether there exists a solution $\varphi\in\mathcal{S}\cap\mathcal{D}$.

Figure 1: The concluding-remarks question on PDF page 8 of arXiv:2505.02098.

2 Definitions and the additional necessary inequality

For $a, b \in \mathbb{D}$, write

$$\rho(a,b) = \left| \frac{a-b}{1-\overline{a}b} \right|$$

for the pseudohyperbolic distance. Let $p_1 = 2, p_2 = 3, \dots$ enumerate the primes, and for $s > 0$ put

$$z(s) = (p_j^{-s})_{j \geq 1} \in \mathbb{B}_{c_0}.$$

The standard Bohr-transform theorem identifies the Banach algebra of bounded Dirichlet series isometrically with $H^\infty(\mathbb{B}_{c_0})$ and carries evaluation at s to evaluation at $z(s)$; see Defant–García–Maestre–Sevilla-Peris [2]. We use the following direct consequence.

Lemma 1 (Bohr-metric contraction)**.** *If $\varphi \in H^\infty(\mathbb{H}_0) \cap \mathcal{D}$ and $\|\varphi\|_\infty \leq 1$, then for $s, t > 0$,*

$$\rho\big(\varphi(s), \varphi(t)\big) \leq \sup_{j \geq 1} \rho(p_j^{-s}, p_j^{-t}). \tag{3}$$

Proof. Let $\Phi \in H^\infty(\mathbb{B}_{c_0})$ be the Bohr lift of φ. For $a \in \mathbb{B}_{c_0}$, the coordinatewise Möbius map

$$\Theta_a(x)_j = \frac{x_j - a_j}{1 - \overline{a_j} x_j}$$

is an automorphism of $\mathbb{B}_{c_0}$ sending a to zero. Compose Φ on the source with $\Theta_{z(s)}^{-1}$ and on the target with the disk automorphism sending $\Phi(z(s))$ to zero. The resulting contractive holomorphic scalar function G on $\mathbb{B}_{c_0}$ satisfies $G(0) = 0$. The Schwarz lemma on the unit ball of a complex Banach space gives $|G(u)| \leq \|u\|_\infty$. Applying it to $u = \Theta_{z(s)}(z(t))$ yields

$$\rho\big(\Phi(z(s)), \Phi(z(t))\big) \leq \|\Theta_{z(s)}(z(t))\|_\infty = \sup_j \rho(p_j^{-s}, p_j^{-t}),$$

which is (3). □

3 Proof intuition

The classical Pick matrix sees only the hyperbolic distance between the two points 1 and 2 in the half-plane. A bounded Dirichlet series has much more structure: under the Bohr lift, those evaluation points become the two infinite-prime vectors $(p^{-1})_p$ and $(p^{-2})_p$. Holomorphic contraction on the c_0 ball therefore imposes a second metric bound. The closest bottleneck is the prime 2, and its exact pseudohyperbolic distance is $2/7$. The classical condition allows target distance all the way up to $1/3$, leaving a nonempty interval of counterexamples.

The same prime-2 coordinate also supplies the sharp positive construction: a disk automorphism applied to 2^{-s} realizes the endpoint $2/7$.

4 Counterexample and exact threshold

Theorem 1. *Let*

$$\lambda_1 = 1, \qquad \lambda_2 = 2, \qquad w_1 = 0, \qquad w_2 = r.$$

For every $r \in \mathbb{D}$ satisfying

$$\frac{2}{7} < |r| < \frac{1}{3}, \tag{4}$$

both matrices (1) and (2) are positive definite (so the latter has rank two), but

$$\mathcal{S} \cap \mathcal{D} = \varnothing.$$

Consequently, the concluding-remarks question of Chavan–Sahu has a negative answer. In particular, one may take $r = 3/10$.

Proof. For these data, the classical matrix is

$$P_{\mathrm{cl}} = \begin{pmatrix} \frac{1}{2} & \frac{1}{3} \\ \frac{1}{3} & \frac{1-|r|^2}{4} \end{pmatrix}, \qquad \det P_{\mathrm{cl}} = \frac{1-9|r|^2}{72}.$$

Thus P_{cl} is positive definite whenever $|r| < 1/3$.

The zeta matrix is

$$P_\zeta = \begin{pmatrix} \zeta(2) & \zeta(3) \\ \zeta(3) & (1-|r|^2)\zeta(4) \end{pmatrix}.$$

The source establishes the strict estimate

$$\frac{\zeta(3)^2}{\zeta(2)\zeta(4)} < \frac{8}{9}. \tag{5}$$

Since $|r| < 1/3$ implies $1 - |r|^2 > 8/9$, we obtain

$$\det P_\zeta = \zeta(2)\zeta(4)\left(1 - |r|^2 - \frac{\zeta(3)^2}{\zeta(2)\zeta(4)}\right) > 0.$$

Hence P_ζ is also positive definite.

Suppose now that some $\varphi \in \mathcal{S} \cap \mathcal{D}$ existed. By the definition of $\mathcal{S}$, it would belong to the closed unit ball of $H^\infty(\mathbb{H}_0)$ and satisfy $\varphi(1) = 0$, $\varphi(2) = r$. Lemma 1 would give

$$\begin{aligned} |r| = \rho(0,r) &\le \sup_{p \text{ prime}} \rho(p^{-1}, p^{-2}) \\ &= \sup_{p \text{ prime}} \frac{p}{p^2+p+1} = \frac{2}{7}. \end{aligned}$$

Indeed, the real function $x/(x^2+x+1)$ has derivative $(1-x^2)/(x^2+x+1)^2 < 0$ for $x > 1$, so the prime supremum occurs at $p = 2$. This contradicts (4).

For the advertised rational choice $r = 3/10$, the classical determinant is exactly

$$\frac{1-(3/10)^2}{8} - \frac{1}{9} = \frac{91}{800} - \frac{1}{9} = \frac{19}{7200} > 0,$$

and (5) gives the strict positivity of P_ζ. $\square$

Corollary 1 (Sharpness on the two-point family)**.** *For $r \in \mathbb{D}$, there is a $\varphi \in H^\infty(\mathbb{H}_0) \cap \mathcal{D}$ with $\|\varphi\|_\infty \le 1$, $\varphi(1) = 0$, and $\varphi(2) = r$ if and only if $|r| \le 2/7$.*

Proof. Necessity is the calculation in Theorem 1. For sufficiency, let

$$B(z) = \frac{\frac{1}{2} - z}{1 - \frac{1}{2}z}.$$

This is a disk automorphism satisfying $B(1/2) = 0$ and $B(1/4) = 2/7$. If $|r| \le 2/7$, define

$$\varphi_r(s) = \frac{7r}{2}\, B(2^{-s}).$$

Then $\|\varphi_r\|_\infty \le 1$ on $\mathbb{H}_0$ and the two required values hold. Moreover,

$$B(z) = \frac{1}{2} - 3\sum_{k=1}^{\infty} \frac{z^k}{2^{k+1}},$$

so $B(2^{-s})$ is a Dirichlet series (supported on powers of 2) converging absolutely throughout $\mathbb{H}_0$. Thus $\varphi_r \in \mathcal{D}$. $\square$

5 Verification and scope

The included script `code/verify_two_point.py` checks the exact rational determinant, evaluates the zeta ratio at 80 decimal digits, verifies the first 25 prime-coordinate distances, and samples the explicit disk automorphism. The calculation is only a sanity check; the proof above is analytic. Full commands and outputs are recorded in `verification.md`.

The counterexample answers the source question exactly as asked. It does not claim that conditions (1)–(2) are never sufficient for other configurations. Corollary 1 is an exact classification only for the displayed two-point family.

6 Bounded novelty check

The run's cheap indexes were searched by arXiv id, title, author names, and the core interpolation terms. Bounded web/arXiv searches through 13 August 2026 used the exact concluding-question phrase, combinations of "$\mathcal{S} \cap \mathcal{D}$", "Bohr transform", "bounded Dirichlet series", the author/title data, and the thresholds 2/7 and 1/3. They located the source and standard Bohr-transform references but no later answer or matching counterexample. Details are in `novelty_search.md`. This is a bounded search, not an exhaustive novelty or priority claim.

7 Human-review recommendation

This packet is suitable for mathematical review as a candidate full negative answer. The highest-value check is that the standard Bohr-transform isometry applies to the source's precise class $H^\infty(\mathbb{H}_0) \cap \mathcal{D}$. Once that standard identification is accepted, the coordinatewise automorphism proof of Lemma 1, both strict Pick-matrix inequalities, and the sharp explicit interpolant are elementary.

References

[1] S. Chavan and C. K. Sahu, *Nevanlinna–Pick interpolation in the right half-plane*, arXiv:2505.02098 (2025).

[2] A. Defant, D. García, M. Maestre, and P. Sevilla-Peris, *Dirichlet Series and Holomorphic Functions in High Dimensions*, New Mathematical Monographs 37, Cambridge University Press, 2019.